



DATA SCIENCE

PROJECT 1: PRODUCT SALES INSIGHTS

Version: 5

Before you start

This project will form the first part of your portfolio that will showcase your talents as a data scientist to potential employers. Remember that you are not the only one attempting this project and so this is your chance to prove that you can supply a better thought out, better looking and easier to use product than anyone else.

Before launching yourself into developing your solution give yourself time to absorb this document; print it out, underline the passages that you are unsure of, research them and prototype the techniques that are required.

Please contact me by email should you require help, guidance or clarification.

1. Overview

A large retailer with 45 nationwide locations has data on sales of 111 weather-sensitive products. The management team would like a data scientist to analyse their data to answer their question “how does weather impact sales of the top three best selling products?” They will use your findings to inform product purchasing. Management would like you to help them with supporting visualizations and source code which answers their business question (refer to the “4. Outputs” section for details).

For example, your findings may indicate that product #123 has higher sales during hot summer weather so management should ensure there is more stock during heatwaves. Or product #456 has more sales during winter storms so management should ensure there is more stock available during forecast severe winter weather.

Your key analysis tasks are to prepare the data, build a model or models on the data which predict how weather influences product sales for the top three best selling products, and use the model outputs to generate business insights. Your model or models should focus on interpretability rather than accuracy, using a simple predictive modelling technique such as a regression rather than a more complex one.

You are expected to use SQL for data preparation and R for analysis. You are not constrained by the contents of the courses, and you may employ any suitable approaches, modelling techniques, packages and libraries to achieve the end result.

2. Minimum skill set

The following skills are required to complete the project:

- SQL
 - Join tables
 - Sort and filter data
- R
 - Data cleaning
 - Feature creation
 - Predictive modelling
 - Data visualization

3. Additional resources

R with RStudio IDE can be used for the analysis and also to connect to SQL and query the SQL data: Download and install R: <https://cran.rstudio.com/> and RStudio IDE: <https://posit.co/download/rstudio-desktop/>.

The data is available from: <https://relational-data.org/dataset/Walmart>. Please refer to the below “Appendix: Data Details” section for the database schema and data dictionary.

Follow the connection instructions on the webpage to establish the SQL data connection. The R packages “DBI” (<https://cran.r-project.org/web/packages/DBI/index.html>) and “RMariaDB” (<https://cran.r-project.org/web/packages/RMariaDB/index.html>) can be used to query the SQL database in R:

```
library("DBI")
library("RMariaDB")
con <- dbConnect(RMariaDB::MariaDB(), dbname="Walmart",
                 host="db.relational-data.org",
                 port="3306", username="guest",
                 password="relational") # connect to the DB
dbListTables(con) # list database tables
res <- dbGetQuery(con, "SELECT * FROM station") # example query
dbFetch(res) # load query results into R dataframe "res"
dbDisconnect(con) # close the connection
```

If the data is not available from the website, you may load it from the supplied SQLite database: “project1_raw_data.db” (see “Appendix: Load SQLite Database in R” for sample code)

Data visualization for the final report may be done as static graphs created in R, for example with the ggplot2 package (<https://ggplot2.tidyverse.org/>) or as an interactive dashboard, for example with Tableau (<https://www.tableau.com/en-gb/products/public>) or Shiny in R (<https://shiny.posit.co/>), though you are not limited to these tools.

The following additional R packages may be useful, though you are not limited to these:

- dplyr (<https://dplyr.tidyverse.org/>): functions for data manipulation
- lubridate (<https://lubridate.tidyverse.org/>): functions to create features from date-time data
- skimr (<https://cran.r-project.org/web/packages/skimr/vignettes/skimr.html>): provide summary statistics on dataframes
- rsample (<https://rsample.tidymodels.org/>): create train-test-validation splits on data
- yardstick (<https://yardstick.tidymodels.org/>): model performance measures

4. Outputs

A written answer sheet (text, plus 3-6 supporting visualizations) which addresses the business question “how does weather impact sales of the top three best selling products?”, plus complete source code.

The visualizations may be static graphs included in the body of the report or a separate interactive dashboard produced by, for example, Tableau or R Shiny. If you produce an interactive dashboard, you should include a link to the published dashboard as well as the source code used to create the dashboard (if relevant).

4.1. Section A: SQL

1. Write a query to find the **top 3 products** with the highest total sales (based on **units** sold) from the **train** table.
2. Using the **key** table, write a query to **join the sales data (train) with the correct weather station** using **store_nbr**.
3. Write a query to return **daily sales and average temperature (tavg)** for one of the top 3 products.

4.2. Section B: R Programming

1. Load the joined dataset into R using **DBI** and **RSQLite**; and convert date columns to **Date** format.
2. Clean the data by:
 - Handling missing weather values (e.g., **preciptotal**, **tavg**).
 - Explain how you dealt with special characters like "T" or empty strings.
 - Avoid dropping rows blindly — consider **imputation** where appropriate (e.g., mean/median for **tavg**).
 - Removing or flagging outliers in **units** sold.
 - **Do not rely solely on IQR if the data is zero-inflated** (e.g., many zeros in units).
 - Justify your approach clearly (percentile trimming, log transform, domain-informed thresholds, etc.).
3. Create at least **two new features** (and up to several if helpful) that could improve your model. (E.g., **is_weekend**, **is_rainy_day** from **codesum**)
 - **Explain why you created each feature** and how it may influence product sales.
4. Build multiple **predictive models (e.g., linear regression, decision tree or other interpretable model)** to predict **units** sold is affected by weather-related features.
 - You may choose to build a separate model for each product or include product as a feature in a combined model.
 - Focus on interpretability over complexity — models should help explain how weather affects sales.

- If you build multiple models, compare them using simple metrics (e.g., R^2 , RMSE, MAE etc.) and explain which is most appropriate and why.
 - Clearly **state which model performs better** and **why** (e.g., better generalisation, interpretability, performance on specific products).
- **Interpret key model outputs:**
 - For Linear Regression, explain what each coefficient means in the context of sales (e.g., how much do units sold increase per 1°F rise in temperature?).
 - For Decision Trees, include feature importance or a visualisation of the tree.

4.3. Section C: Interpretation & Visualization

1. Write **3 short paragraphs** (one per product), summarizing how weather impacts sales for each.

- Which weather variable has the strongest influence on sales?
- Use evidence from your model output (e.g., R^2 scores, coefficients, variable importance) to support your analysis.
- Is the relationship positive or negative?

Be concise but insightful. Your explanation should connect back to the real-world business context

2. Visualizations:

Produce **3–6 visualizations** that support your analysis and illustrate your interpretations. These may include:

- Scatter plots showing correlation (e.g., temperature vs. sales)
- Line plots over time
- Bar charts grouped by weather events
- Optionally: embed or link to an interactive dashboard (e.g., via R Shiny or Tableau Public)
- All plots must:
 - Include titles, labels, and legends.
 - Be accompanied by 1–2 sentences explaining what the chart shows.

3. Business Recommendations:

Provide **3–6 actionable business recommendations** based on your findings. These should:

- Be **based on evidence** from your model interpretations or visualisations.
- Be **product-specific** — at least **one or two per product**.
- Link to **weather-based insights** (e.g., “Increase stock of Product 9 in summer months due to temperature-driven demand rise.”)
- Be practical and show **understanding**

Note: Good recommendations are **specific, justified, and data-driven** — not general or vague

The source code should be fully reproducible, e.g. another data scientist with access to the data could run the code from end-to-end to produce the results shown in the report/answer sheet. It should contain the following sections:

- Data Preparation (SQL): Join the weather and the station data, filter it to the top three best-selling products.
- Data Cleaning and Feature Creation (R): Check for data inconsistencies and missing data, address as needed. Create any features needed for modelling.
- Predictive Model(s) (R): Model or models which predict product sales based on weather features for the top three best selling products. You may also include non-weather features if they improve model performance. Model(s) should focus on simplicity and interpretability rather than accuracy. Use data science best practice for model building, such as a train-test-validate split.
- Model Outputs (R): For each model built, output the model summary and an appropriate performance measure (such as R^2 or root mean square error) on a validation split at a minimum. Optionally, include any additional outputs to support your written findings.
- Visualizations (R or dashboarding tool): source code for static visualizations or a programmed dashboard. If a non-programming tool such as Tableau was used, include a short paragraph on how you created the dashboard.

5. Project approach

The following approach is suggested to produce the project outputs required from the data, but the final decision is up to you. Start by planning what you would like your final report and visualization to contain and then plan the work required to create those from the data:

- Load the SQL database
- Use SQL to prepare the analysis dataset (joining and filtering)
- Load the analysis dataset into R
- Data cleaning and feature creation in R
- Create a predictive model or models in R
- Use the model outputs to get insights in R
- Create the visualizations (R or dashboarding tool) and write the report

Appendix: Data Details

Data Dictionary:

“train” table:

- date – date of the sales
- store_nbr - an id representing one of the 45 stores
- item_nbr - an id representing one of the 111 products
- units - the quantity sold of an item on a given day

“key” table:

- store_nbr - an id representing one of the 45 stores
- station_nbr - an id representing one of 20 weather stations

“station” table:

- station_nbr - an id representing one of 20 weather stations

“weather” table:

- station_nbr - an id representing one of 20 weather stations
- date – date of the weather
- tmax – maximum temperature (degrees Fahrenheit)
- tmin – minimum temperature (degrees Fahrenheit)
- tavg – average temperature (degrees Fahrenheit)
- depart – departure from normal (degrees Fahrenheit)
- dewpoint – average dew point (degrees Fahrenheit)
- wetbulb – average wet bulb (degrees Fahrenheit)
- heat – heating degree days (base 65 degrees Fahrenheit, season begins with July)
- cool – cooling degree days (base 65 degrees Fahrenheit, season begins with January)
- sunrise – calculated time of sunrise
- sunset – calculated time of sunset
- codesum – significant weather event code
 - +fc: tornado/waterspout
 - fc: funnel cloud
 - ts: thunderstorm
 - gr: hail

train	
date	date
store_nbr	int
item_nbr	int
units	int

key	
store_nbr	int
station_nbr	int

weather	
station_nbr	int
date	date
tmax	int
tmin	int
tavg	int
depart	int
dewpoint	int
wetbulb	int
heat	int
cool	int
sunrise	time
sunset	time
codesum	varchar
snowfall	float
preciptotal	float
stnpressure	float
sealevel	float
resultspeed	float
resultdir	int
avgspeed	float

station	
station_nbr	int

- ra: rain
- dz: drizzle
- sn: snow
- sg: snow grains
- gs: small hail &/or snow pellets
- pl: ice pellets
- ic: ice crystals
- fg+: heavy fog (fg & le.25 miles visibility)
- fg: fog
- br: mist
- up: unknown precipitation
- hz: haze
- fu: smoke
- va: volcanic ash
- du: widespread dust
- ds: duststorm
- po: sand/dust whirls
- sa: sand
- ss: sandstorm
- py: spray
- sq: squall
- dr: low drifting
- sh: shower
- fz: freezing
- mi: shallow
- pr: partial
- bc: patches
- bl: blowing
- vc: vicinity
- -: light
- +: heavy
- no sign: moderate
- snowfall – snowfall (inches and tenths) in the 24 hours ending at 2400 LST, T = trace (note not all stations report snow totals)
- preciptotal – water equivalent (inches and hundredths) of precipitation (rainfall and melted snow) in the 24 hours ending at 2400 LST, T = trace
- stnpressure – average station atmospheric pressure in inches of Hg
- sealevel – average sea level atmospheric pressure in inches of Hg
- resultsspeed – wind speed in miles per hour
- resultdir – wind direction in degrees to tens of degrees
- avgspeed – average wind speed in miles per hour

Appendix: Load SQLite Database in R

The R packages “DBI” (<https://cran.r-project.org/web/packages/DBI/index.html>) and “RSQLite” (<https://cran.r-project.org/web/packages/RSQLite/vignettes/RSQLite.html>) can be used to query the supplied SQLite database file “project1_raw_data.db” in R:

```
library("DBI")
library("RSQLite")
con <- dbConnect(RSQLite::SQLite(),
                  "./project1_raw_data.db") # connect to the DB
dbListTables(con) # list database tables
res <- dbGetQuery(con, "SELECT * FROM station") # example query
dbFetch(res) # load query results into R dataframe “res”
dbDisconnect(con) # close the connection
```